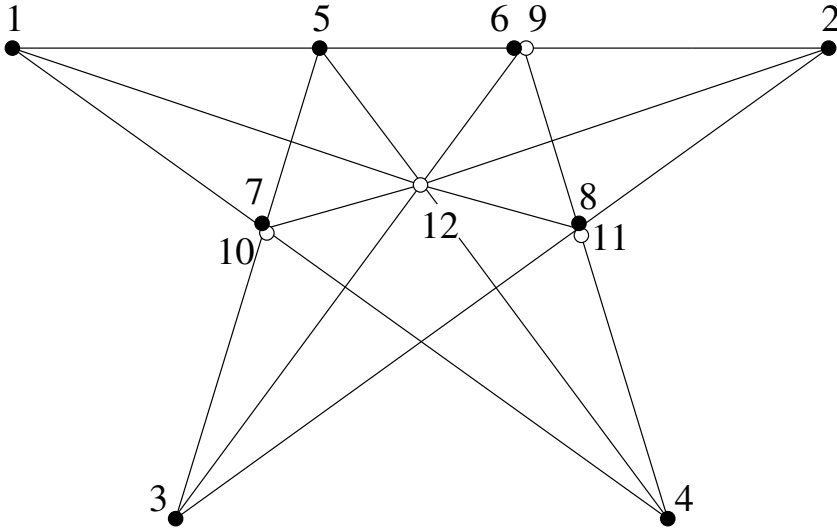


2. G is the Gale diagram of an 8-dimensional polytope with 12 vertices (check this!); a polytope represented by G cannot have all coordinates rational.



3. Consider any spanning configuration G' of 12 signed points in the plane (an affine Gale diagram in $\mathbb{R}^2$). If G' has the same positive circuits as G , then the three pairs of points that coincide in G have to coincide in G' as well, and the triples and quadruples that are collinear in G have to be collinear in G' as well, because they are all positive vectors (unions of positive circuits).

Thus also G' cannot be realized with rational coordinates.

Thus G is the Gale diagram of a nonrational polytope P . If P' is a polytope that is combinatorially equivalent to P , then its Gale diagram G' has the same positive circuits as the Gale diagram G , hence with part 3. above G' and P' cannot be rational either.

(b) Facets of 4-Polytopes Cannot be Prescribed

Perles apparently first observed that the shape of a facet of a d -polytope cannot in general be prescribed; see Grünbaum [252, p. 96, Ex. 3]. Kleinschmidt [332] finally constructed a 4-polytope with 8 vertices for which the shape of a facet cannot be prescribed — this is the smallest dimension and the minimal number of vertices for such an example, because all facets can be prescribed for 3-polytopes, and for all d -polytope with at most $d+3$ vertices. With $d = 4$ and $n = 8$, Kleinschmidt's polytope can be constructed

as a 2-dimensional affine Gale diagram (Exercise 6.18). It has the special property that all facets except the “bad” one are simplicial.

In Example 5.11 we saw “Barnette’s example,” which has a minimal number of facets, namely 7. In fact, if P is a prism over a square pyramid, then the shape of its cubical facet cannot be prescribed. The square pyramid Pyr_3 is isomorphic to its polar. Thus we get that P^Δ , a bipyramid over a square pyramid, is a 4-polytope with 7 vertices for which a vertex figure cannot be prescribed.

The following constructs an (equivalent) Gale diagram description of Barnette’s example [47], which Sturmfels [532] found independently of Barnette’s work.

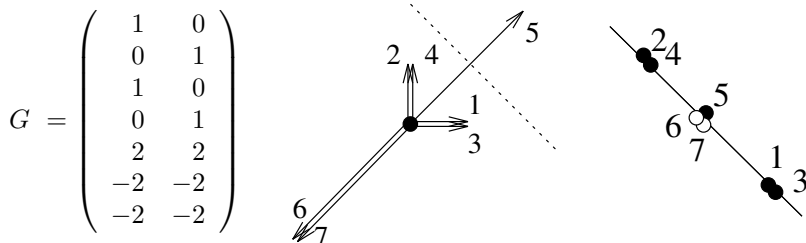
Example 6.22. [532, Prop. 5.1]

There is a 4-polytope P with 7 facets for which the shape of a facet cannot be prescribed.

For this, let P^Δ be the bipyramid over a square pyramid, as given by

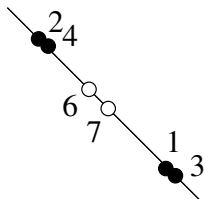
$$V = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 \end{pmatrix} \sim \begin{pmatrix} 0 & 0 & 2 & 2 & 0 & 0 & 1 \\ 1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -2 & -2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 2 & 0 & 1 & 0 \end{pmatrix},$$

from which we read off a Gale diagram



Now we observe that the vertex figure at the vertex 5 is an octahedron. Its Gale diagram is obtained by deleting the point 5 from the diagram: it corresponds to a regular octahedron (compare to Example 6.18).

Now for P^Δ we see that $5\bar{6}$ and $5\bar{7}$ are positive cofacets, which requires that the points 6 and 7 coincide on the affine Gale diagram of the vertex figure $P^\Delta/5$. Thus, if we start with the nonregular octahedron with Gale diagram



then this is not the vertex figure of a 4-polytope that is combinatorially equivalent to P^Δ .

(c) 2-Faces of 5-Polytopes Cannot be Prescribed

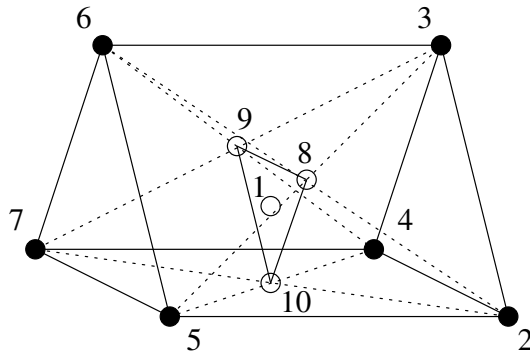
We have now seen the second proof that the shape of a facet cannot be prescribed for 4-polytopes, so we'll have to start and accept it.

But perhaps this was the wrong generalization of the 3-dimensional theorem that a facet can be prescribed. What about prescribing the shape of a 2-face for a d -polytope? The case $d = 4$ was an open problem in the first edition of this book (Problem 6.11*), now you can find it as Exercise 6.11. For $d = 5$ we have the following counterexample from the first edition. It is the type of analysis that Gale diagrams “were made for”: however, the construction of this example may have been the first time that a 3-dimensional affine Gale diagram was seriously used.

Example 6.23. *There is a 5-polytope P with 10 facets and 12 vertices, for which the shape of a 2-face cannot be prescribed.*

To prove this, we construct the polar polytope $Q := P^\Delta$, a 5-polytope with 10 vertices and 12 facets, and verify that for one contraction (face figure, see Exercise 2.9) of a 2-face, the shape cannot be prescribed.

For this consider the signed point configuration given by the vertices of a triangular prism as positive points, labeled $2, 3, \dots, 7$, the centers of the square facets of the prism as negative points $8, 9, 10$, and the center of the whole prism as another negative point 1 . See the figure for how we label this.



It is easy to give coordinates. In fact, the corresponding vector configuration in $\mathbb{R}^4$ could be taken to be

$$\begin{pmatrix} -1 & 1 & 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 \\ -1 & 2 & 2 & 2 & 0 & 0 & 0 & -1 & -1 & -1 \\ -2 & 0 & 0 & 6 & 0 & 0 & 6 & 0 & -3 & -3 \\ -2 & 0 & 6 & 0 & 0 & 6 & 0 & -3 & -3 & 0 \end{pmatrix}.$$